

- **29.** Show that if m and n are even positive integers, the crossing number of $K_{m,n}$ is less than or equal to $mn(m-2)(n-2)/16$. [Hint: Place m vertices along the x -axis so that they are equally spaced and symmetric about the origin and place n vertices along the y -axis so that they are equally spaced and symmetric about the origin. Now connect each of the m vertices on the x -axis to each of the vertices on the y -axis and count the crossings.]

The **thickness** of a simple graph G is the smallest number of planar subgraphs of G that have G as their union.

- 30.** Show that $K_{3,3}$ has 2 as its thickness.
- *31.** Find the thickness of the graphs in Exercise 27.
- 32.** Show that if G is a connected simple graph with v vertices and e edges, where $v \geq 3$, then the thickness of G is at least $\lceil e/(3v-6) \rceil$.
- *33.** Use Exercise 32 to show that the thickness of K_n is at least $\lfloor (n+7)/6 \rfloor$ whenever n is a positive integer.
- 34.** Show that if G is a connected simple graph with v vertices and e edges, where $v \geq 3$, and no circuits of length three, then the thickness of G is at least $\lceil e/(2v-4) \rceil$.
- 35.** Use Exercise 34 to show that the thickness of $K_{m,n}$, where m and n are not both 1, is at least $\lceil mn/(2m+2n-4) \rceil$ whenever m and n are positive integers.
- *36.** Draw K_5 on the surface of a torus (a doughnut-shaped solid) so that no edges cross.
- *37.** Draw $K_{3,3}$ on the surface of a torus so that no edges cross.

10.8 Graph Coloring

10.8.1 Introduction

Links

Problems related to the coloring of maps of regions, such as maps of parts of the world, have generated many results in graph theory. When a map* is colored, two regions with a common border are customarily assigned different colors. One way to ensure that two adjacent regions never have the same color is to use a different color for each region. However, this is inefficient, and on maps with many regions it would be hard to distinguish similar colors. Instead, a small number of colors should be used whenever possible. Consider the problem of determining the least number of colors that can be used to color a map so that adjacent regions never have the same color. For instance, for the map shown on the left in Figure 1, four colors suffice, but three colors are not enough. (The reader should check this.) In the map on the right in Figure 1, three colors are sufficient (but two are not).

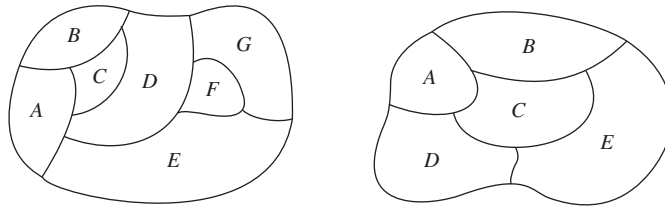


FIGURE 1 Two maps.

Each map in the plane can be represented by a graph. To set up this correspondence, each region of the map is represented by a vertex. Edges connect two vertices if the regions represented by these vertices have a common border. Two regions that touch at only one point are not considered adjacent. The resulting graph is called the **dual graph** of the map. By the way in which dual graphs of maps are constructed, it is clear that any map in the plane has a planar dual graph. Figure 2 displays the dual graphs that correspond to the maps shown in Figure 1.

The problem of coloring the regions of a map is equivalent to the problem of coloring the vertices of the dual graph so that no two adjacent vertices in this graph have the same color. We now define a graph coloring.

*We will assume that all regions in a map are connected. This eliminates any problems presented by such geographical entities as Michigan.

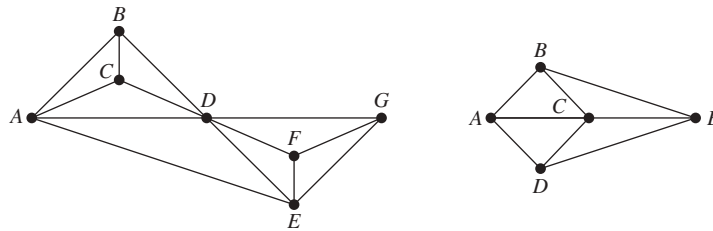


FIGURE 2 Dual graphs of the maps in Figure 1.

Definition 1

A *coloring* of a simple graph is the assignment of a color to each vertex of the graph so that no two adjacent vertices are assigned the same color.

A graph can be colored by assigning a different color to each of its vertices. However, for most graphs a coloring can be found that uses fewer colors than the number of vertices in the graph. What is the least number of colors necessary?

Definition 2

The *chromatic number* of a graph is the least number of colors needed for a coloring of this graph. The chromatic number of a graph G is denoted by $\chi(G)$. (Here χ is the Greek letter *chi*.)

Note that asking for the chromatic number of a planar graph is the same as asking for the minimum number of colors required to color a planar map so that no two adjacent regions are assigned the same color. This question has been studied for more than 100 years. The answer is provided by one of the most famous theorems in mathematics.

THEOREM 1

THE FOUR COLOR THEOREM The chromatic number of a planar graph is no greater than four.



The four color theorem was originally posed as a conjecture in the 1850s. It was finally proved by the American mathematicians Kenneth Appel and Wolfgang Haken in 1976. Prior to 1976, many incorrect proofs were published, often with hard-to-find errors. In addition, many futile attempts were made to construct counterexamples by drawing maps that require more than four colors. (Proving the five color theorem is not that difficult; see Exercise 36.)



Perhaps the most notorious fallacious proof in all of mathematics is the incorrect proof of the four color theorem published in 1879 by a London barrister and amateur mathematician, Alfred Kempe. Mathematicians accepted his proof as correct until 1890, when Percy Heawood found an error that made Kempe's argument incomplete. However, Kempe's line of reasoning turned out to be the basis of the successful proof given by Appel and Haken. Their proof relies on a careful case-by-case analysis carried out by computer. They showed that if the four color theorem were false, there would have to be a counterexample of one of approximately 2000 different



Courtesy of London
Mathematical Society

ALFRED BRAY KEMPE (1849–1922) Kempe was a barrister and a leading authority on ecclesiastical law. However, having studied mathematics at Cambridge University, he retained his interest in it, and later in life he devoted considerable time to mathematical research. Kempe made contributions to kinematics, the branch of mathematics dealing with motion, and to mathematical logic. However, Kempe is best remembered for his fallacious proof of the four color theorem.

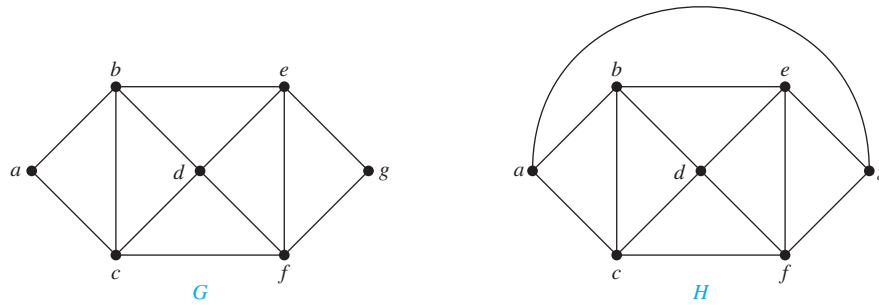


FIGURE 3 The simple graphs G and H .

types, and they then showed that none of these types exists. They used over 1000 hours of computer time in their proof. This proof generated a large amount of controversy, because computers played such an important role in it. For example, could there be an error in a computer program that led to incorrect results? Was their argument really a proof if it depended on what could be unreliable computer output? Since their proof appeared, simpler proofs that rely on checking fewer types of possible counterexamples have been found and a proof using an automated proof system has been created. However, no proof not relying on a computer has yet been found.

Note that the four color theorem applies only to planar graphs. Nonplanar graphs can have arbitrarily large chromatic numbers, as will be shown in Example 2.

Two things are required to show that the chromatic number of a graph is k . First, we must show that the graph can be colored with k colors. This can be done by constructing such a coloring. Second, we must show that the graph cannot be colored using fewer than k colors. Examples 1–4 illustrate how chromatic numbers can be found.

EXAMPLE 1 What are the chromatic numbers of the graphs G and H shown in Figure 3?

Extra Examples ➤

Solution: The chromatic number of G is at least three, because the vertices a , b , and c must be assigned different colors. To see if G can be colored with three colors, assign red to a , blue to b , and green to c . Then, d can (and must) be colored red because it is adjacent to b and c . Furthermore, e can (and must) be colored green because it is adjacent only to vertices colored red and blue, and f can (and must) be colored blue because it is adjacent only to vertices colored red and green. Finally, g can (and must) be colored red because it is adjacent only to vertices colored blue and green. This produces a coloring of G using exactly three colors. Figure 4 displays such a coloring.

The graph H is made up of the graph G with an edge connecting a and g . Any attempt to color H using three colors must follow the same reasoning as that used to color G , except at the last stage, when all vertices other than g have been colored. Then, because g is adjacent (in H) to vertices colored red, blue, and green, a fourth color, say brown, needs to be used. Hence, H has a chromatic number equal to 4. A coloring of H is shown in Figure 4. ◀

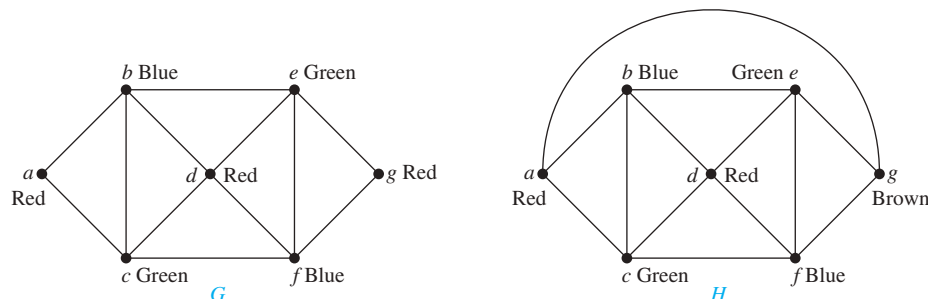
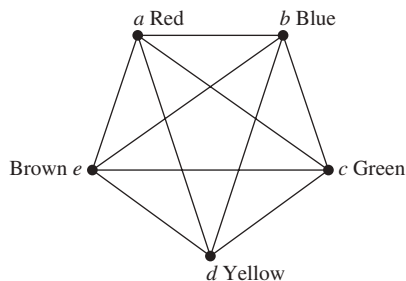
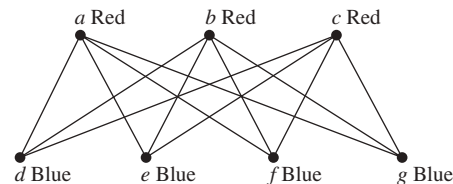


FIGURE 4 Colorings of the graphs G and H .

FIGURE 5 A coloring of K_5 .FIGURE 6 A coloring of $K_{3,4}$.

EXAMPLE 2 What is the chromatic number of K_n ?

Solution: A coloring of K_n can be constructed using n colors by assigning a different color to each vertex. Is there a coloring using fewer colors? The answer is no. No two vertices can be assigned the same color, because every two vertices of this graph are adjacent. Hence, the chromatic number of K_n is n . That is, $\chi(K_n) = n$. (Recall that K_n is not planar when $n \geq 5$, so this result does not contradict the four color theorem.) A coloring of K_5 using five colors is shown in Figure 5. ▶

EXAMPLE 3 What is the chromatic number of the complete bipartite graph $K_{m,n}$, where m and n are positive integers?

Solution: The number of colors needed may seem to depend on m and n . However, as Theorem 4 in Section 10.2 tells us, only two colors are needed, because $K_{m,n}$ is a bipartite graph. Hence, $\chi(K_{m,n}) = 2$. This means that we can color the set of m vertices with one color and the set of n vertices with a second color. Because edges connect only a vertex from the set of m vertices and a vertex from the set of n vertices, no two adjacent vertices have the same color. A coloring of $K_{3,4}$ with two colors is displayed in Figure 6. ▶

EXAMPLE 4 What is the chromatic number of the graph C_n , where $n \geq 3$? (Recall that C_n is the cycle with n vertices.)

Solution: We will first consider some individual cases. To begin, let $n = 6$. Pick a vertex and color it red. Proceed clockwise in the planar depiction of C_6 shown in Figure 7. It is necessary to assign a second color, say blue, to the next vertex reached. Continue in the clockwise direction;

HISTORICAL NOTE In 1852, an ex-student of Augustus De Morgan, Francis Guthrie, noticed that the counties in England could be colored using four colors so that no adjacent counties were assigned the same color. On this evidence, he conjectured that the four color theorem was true. Francis told his brother Frederick, at that time a student of De Morgan, about this problem. Frederick in turn asked his teacher De Morgan about his brother's conjecture. De Morgan was extremely interested in this problem and publicized it throughout the mathematical community. In fact, the first written reference to the conjecture can be found in a letter from De Morgan to Sir William Rowan Hamilton. Although De Morgan thought Hamilton would be interested in this problem, Hamilton apparently was not interested in it, because it had nothing to do with quaternions.



HISTORICAL NOTE Although a simpler proof of the four color theorem was found by Robertson, Sanders, Seymour, and Thomas in 1996, reducing the computational part of the proof to examining 633 configurations, no proof that does not rely on extensive computation has yet been found.

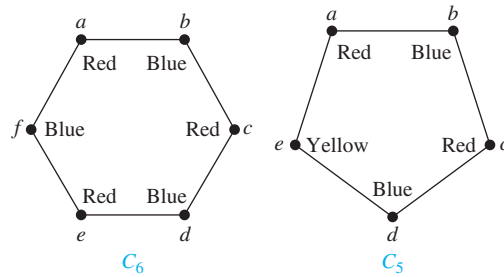


FIGURE 7 Colorings of C_5 and C_6 .

the third vertex can be colored red, the fourth vertex blue, and the fifth vertex red. Finally, the sixth vertex, which is adjacent to the first, can be colored blue. Hence, the chromatic number of C_6 is 2. Figure 7 displays the coloring constructed here.

Next, let $n = 5$ and consider C_5 . Pick a vertex and color it red. Proceeding clockwise, it is necessary to assign a second color, say blue, to the next vertex reached. Continuing in the clockwise direction, the third vertex can be colored red, and the fourth vertex can be colored blue. The fifth vertex cannot be colored either red or blue, because it is adjacent to the fourth vertex and the first vertex. Consequently, a third color is required for this vertex. Thus, the chromatic number of C_5 is 3. A coloring of C_5 using three colors is displayed in Figure 7.

In general, two colors are needed to color C_n when n is even. To construct such a coloring, simply pick a vertex and color it red. Proceed around the graph in a clockwise direction (using a planar representation of the graph) coloring the second vertex blue, the third vertex red, and so on. The n th vertex can be colored blue, because the two vertices adjacent to it, namely the $(n - 1)$ st and the first vertices, are both colored red.

When n is odd and $n > 1$, the chromatic number of C_n is 3. To see this, pick an initial vertex. To use only two colors, it is necessary to alternate colors as the graph is traversed in a clockwise direction. However, the n th vertex reached is adjacent to two vertices of different colors, namely, the first and $(n - 1)$ st. Hence, a third color must be used.

We have shown that $\chi(C_n) = 2$ if n is an even positive integer with $n \geq 4$ and $\chi(C_n) = 3$ if n is an odd positive integer with $n \geq 3$. ◀

Links

The best algorithms known for finding the chromatic number of a graph have exponential worst-case time complexity (in the number of vertices of the graph). Even the problem of finding an approximation to the chromatic number of a graph is difficult. It has been shown that if there were an algorithm with polynomial worst-case time complexity that could approximate the chromatic number of a graph up to a factor of 2 (that is, construct a bound that was no more than double the chromatic number of the graph), then an algorithm with polynomial worst-case time complexity for finding the chromatic number of the graph would also exist.

10.8.2 Applications of Graph Colorings

Graph coloring has a variety of applications to problems involving scheduling and assignments. (Note that because no efficient algorithm is known for graph coloring, this does not lead to efficient algorithms for scheduling and assignments.) Examples of such applications will be given here. The first application deals with the scheduling of final exams.

EXAMPLE 5 Scheduling Final Exams How can the final exams at a university be scheduled so that no student has two exams at the same time?

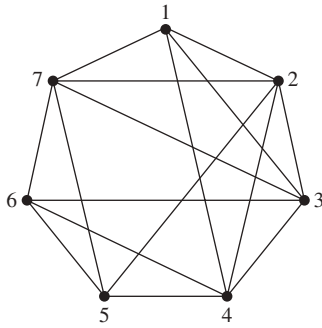


FIGURE 8 The graph representing the scheduling of final exams.

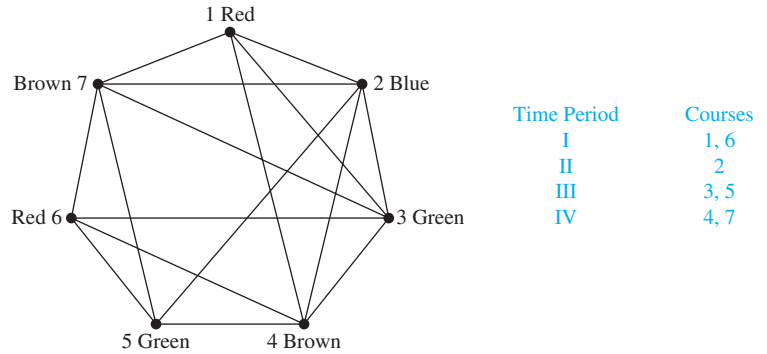


FIGURE 9 Using a coloring to schedule final exams.

Solution: This scheduling problem can be solved using a graph model, with vertices representing courses and with an edge between two vertices if there is a common student in the courses they represent. Each time slot for a final exam is represented by a different color. A scheduling of the exams corresponds to a coloring of the associated graph.

For instance, suppose there are seven finals to be scheduled. Suppose the courses are numbered 1 through 7. Suppose that the following pairs of courses have common students: 1 and 2, 1 and 3, 1 and 4, 1 and 7, 2 and 3, 2 and 4, 2 and 5, 2 and 7, 3 and 4, 3 and 6, 3 and 7, 4 and 5, 4 and 6, 5 and 6, 5 and 7, and 6 and 7. In Figure 8 the graph associated with this set of classes is shown. A scheduling consists of a coloring of this graph.

Because the chromatic number of this graph is 4 (the reader should verify this), four time slots are needed. A coloring of the graph using four colors and the associated schedule are shown in Figure 9.

Now consider an application to the assignment of television channels.

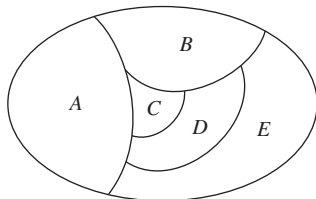
EXAMPLE 6 Frequency Assignments Television channels 2 through 13 are assigned to stations in North America so that no two stations within 150 miles can operate on the same channel. How can the assignment of channels be modeled by graph coloring?

Solution: Construct a graph by assigning a vertex to each station. Two vertices are connected by an edge if they are located within 150 miles of each other. An assignment of channels corresponds to a coloring of the graph, where each color represents a different channel.

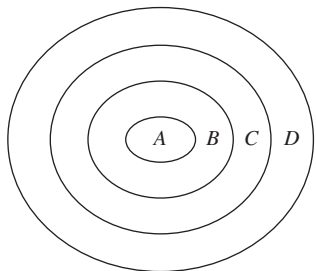
An application of graph coloring to compilers is considered in Example 7.

EXAMPLE 7 Index Registers In efficient compilers the execution of loops is speeded up when frequently used variables are stored temporarily in index registers in the central processing unit, instead of in regular memory. For a given loop, how many index registers are needed? This problem can be addressed using a graph coloring model. To set up the model, let each vertex of a graph represent a variable in the loop. There is an edge between two vertices if the variables they represent must be stored in index registers at the same time during the execution of the loop. Thus, the chromatic number of the graph gives the number of index registers needed, because different registers must be assigned to variables when the vertices representing these variables are adjacent in the graph.

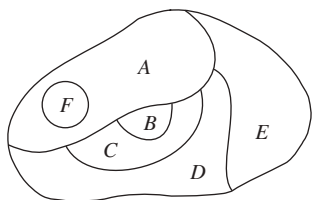
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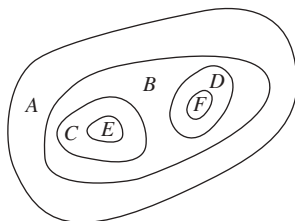
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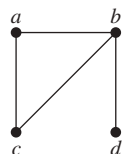
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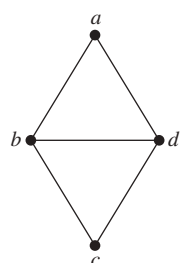
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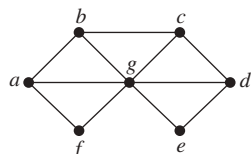
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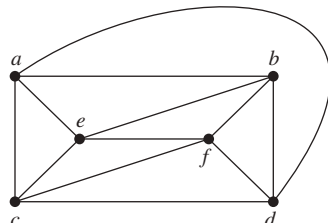
7.



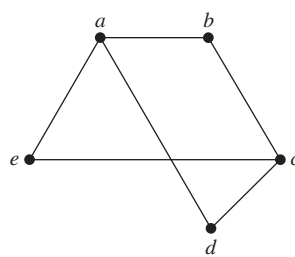
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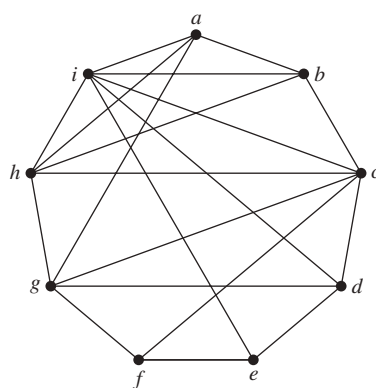
8.



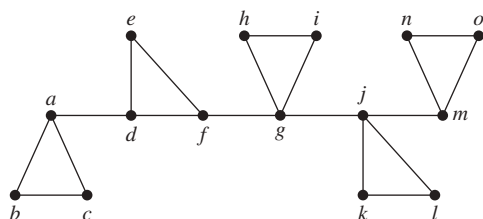
9.



10.



11.



13. Which graphs have a chromatic number of 1?

15. What is the chromatic number of W_n ?

17. Schedule the final exams for Math 115, Math 116, Math 185, Math 195, CS 101, CS 102, CS 273, and CS 473, using the fewest number of different time slots, if there are no students taking both Math 115 and CS 473, both Math 116 and CS 473, both Math 195 and CS 101, both Math 195 and CS 102, both Math 115 and Math 116, both Math 115 and Math 185, and both Math 185 and Math 195, but there are students in every other pair of courses.

18. How many different channels are needed for six stations located at the distances shown in the table, if two stations cannot use the same channel when they are within 150 miles of each other?

	1	2	3	4	5	6
1	—	85	175	200	50	100
2	85	—	125	175	100	160
3	175	125	—	100	200	250
4	200	175	100	—	210	220
5	50	100	200	210	—	100
6	100	160	250	220	100	—

19. The mathematics department has six committees, each meeting once a month. How many different meeting times must be used to ensure that no member is scheduled to attend two meetings at the same time if the committees are $C_1 = \{\text{Arlinghaus, Brand, Zaslavsky}\}$, $C_2 = \{\text{Brand, Lee, Rosen}\}$, $C_3 = \{\text{Arlinghaus, Rosen, Zaslavsky}\}$, $C_4 = \{\text{Lee, Rosen, Zaslavsky}\}$, $C_5 = \{\text{Arlinghaus, Brand}\}$, and $C_6 = \{\text{Brand, Rosen, Zaslavsky}\}$?
20. A zoo wants to set up natural habitats in which to exhibit its animals. Unfortunately, some animals will eat some of the others when given the opportunity. How can a graph model and a coloring be used to determine the number of different habitats needed and the placement of the animals in these habitats?

► An **edge coloring** of a graph is an assignment of colors to edges so that edges incident with a common vertex are assigned different colors. The **edge chromatic number** of a graph is the smallest number of colors that can be used in an edge coloring of the graph. The edge chromatic number of a graph G is denoted by $\chi'(G)$.

21. Find the edge chromatic number of each of the graphs in Exercises 5–11.
22. Suppose that n devices are on a circuit board and that these devices are connected by colored wires. Express the number of colors needed for the wires, in terms of the edge chromatic number of the graph representing this circuit board, under the requirement that the wires leaving a particular device must be different colors. Explain your answer.
23. Find the edge chromatic numbers of
- C_n , where $n \geq 3$.
 - W_n , where $n \geq 3$.

24. Show that the edge chromatic number of a graph must be at least as large as the maximum degree of a vertex of the graph.

25. Show that if G is a graph with n vertices, then no more than $n/2$ edges can be colored the same in an edge coloring of G .

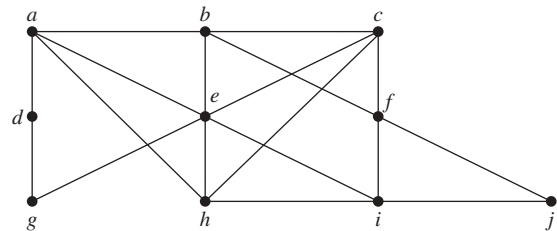
- *26. Find the edge chromatic number of K_n when n is a positive integer.

27. Seven variables occur in a loop of a computer program. The variables and the steps during which they must be stored are t : steps 1 through 6; u : step 2; v : steps 2 through 4; w : steps 1, 3, and 5; x : steps 1 and 6; y : steps 3 through 6; and z : steps 4 and 5. How many different index registers are needed to store these variables during execution?

28. What can be said about the chromatic number of a graph that has K_n as a subgraph?

This algorithm can be used to color a simple graph: First, list the vertices $v_1, v_2, v_3, \dots, v_n$ in order of decreasing degree so that $\deg(v_1) \geq \deg(v_2) \geq \dots \geq \deg(v_n)$. Assign color 1 to v_1 and to the next vertex in the list not adjacent to v_1 (if one exists), and successively to each vertex in the list not adjacent to a vertex already assigned color 1. Then assign color 2 to the first vertex in the list not already colored. Successively assign color 2 to vertices in the list that have not already been colored and are not adjacent to vertices assigned color 2. If uncolored vertices remain, assign color 3 to the first vertex in the list not yet colored, and use color 3 to successively color those vertices not already colored and not adjacent to vertices assigned color 3. Continue this process until all vertices are colored.

29. Construct a coloring of the graph shown using this algorithm.



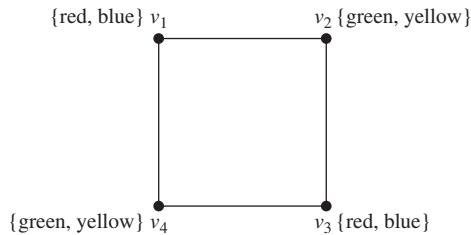
- *30. Use pseudocode to describe this coloring algorithm.
- *31. Show that the coloring produced by this algorithm may use more colors than are necessary to color a graph.

A connected graph G is called **chromatically k -critical** if the chromatic number of G is k , but for every edge of G , the chromatic number of the graph obtained by deleting this edge from G is $k - 1$.

32. Show that C_n is chromatically 3-critical whenever n is an odd positive integer, $n \geq 3$.
33. Show that W_n is chromatically 4-critical whenever n is an odd integer, $n \geq 3$.
34. Show that W_4 is not chromatically 3-critical.

35. Show that if G is a chromatically k -critical graph, then the degree of every vertex of G is at least $k - 1$.

A k -tuple coloring of a graph G is an assignment of a set of k different colors to each of the vertices of G such that no two adjacent vertices are assigned a common color. We denote by $\chi_k(G)$ the smallest positive integer n such that G has a k -tuple coloring using n colors. For example, $\chi_2(C_4) = 4$. To see this, note that using only four colors we can assign two colors to each vertex of C_4 , as illustrated, so that no two adjacent vertices are assigned the same color. Furthermore, no fewer than four colors suffice because the vertices v_1 and v_2 each must be assigned two colors, and a common color cannot be assigned to both v_1 and v_2 . (For more information about k -tuple coloring, see [MiRo91].)



36. Find these values:

- a) $\chi_2(K_3)$ b) $\chi_2(K_4)$ c) $\chi_2(W_4)$
 d) $\chi_2(C_5)$ e) $\chi_2(K_{3,4})$ f) $\chi_3(K_5)$
 *g) $\chi_3(C_5)$ h) $\chi_3(K_{4,5})$

- *37. Let G and H be the graphs displayed in Figure 3. Find

- a) $\chi_2(G)$. b) $\chi_2(H)$.
 c) $\chi_3(G)$. d) $\chi_3(H)$.

38. What is $\chi_k(G)$ if G is a bipartite graph and k is a positive integer?

39. Frequencies for mobile radio (or cellular) telephones are assigned by zones. Each zone is assigned a set of frequencies to be used by vehicles in that zone. The same frequency cannot be used in different zones when interference can occur between telephones in these zones. Explain how a k -tuple coloring can be used to assign k frequencies to each mobile radio zone in a region.

- *40. Show that every planar graph G can be colored using six or fewer colors. [Hint: Use mathematical induction on the number of vertices of the graph. Apply Corollary 2 of Section 10.7 to find a vertex v with $\deg(v) \leq 5$. Consider the subgraph of G obtained by deleting v and all edges incident with it.]

- **41. Show that every planar graph G can be colored using five or fewer colors. [Hint: Use the hint provided for Exercise 40.]

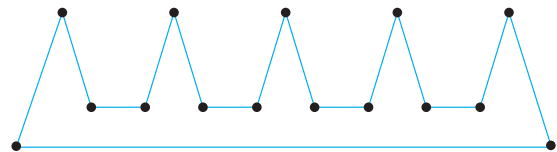
The **art gallery problem** asks how many guards are needed to see all parts of an art gallery, where the gallery is the interior and boundary of a polygon with n sides. To state this problem more precisely, we need some terminology. A point x inside or on the boundary of a simple polygon P **covers** or **sees** a point y inside or on P if all points on the line segment xy are in the interior or on the boundary of P . We say that a set of points is a **guarding set** of a simple polygon P if for every point y inside P or on the boundary of P there is a point x in this guarding set that sees y . Denote by $G(P)$ the minimum number of points needed to guard the simple polygon P . The art gallery problem asks for the function $g(n)$, which is the maximum value of $G(P)$ over all simple polygons with n vertices. That is, $g(n)$ is the minimum positive integer for which it is guaranteed that a simple polygon with n vertices can be guarded with $g(n)$ or fewer guards.

42. Show that $g(3) = 1$ and $g(4) = 1$ by showing that all triangles and quadrilaterals can be guarded using one point.

- *43. Show that $g(5) = 1$. That is, show that all pentagons can be guarded using one point. [Hint: Show that there are either 0, 1, or 2 vertices with an interior angle greater than 180 degrees and that in each case, one guard suffices.]

- *44. Show that $g(6) = 2$ by first using Exercises 42 and 43 as well as Lemma 1 in Section 5.2 to show that $g(6) \leq 2$ and then find a simple hexagon for which two guards are needed.

- *45. Show that $g(n) \geq \lfloor n/3 \rfloor$. [Hint: Consider the polygon with $3k$ vertices that resembles a comb with k prongs, such as the polygon with 15 sides shown here.]



- *46. Solve the art gallery problem by proving the **art gallery theorem**, which states that at most $\lfloor n/3 \rfloor$ guards are needed to guard the interior and boundary of a simple polygon with n vertices. [Hint: Use Theorem 1 in Section 5.2 to triangulate the simple polygon into $n - 2$ triangles. Then show that it is possible to color the vertices of the triangulated polygon using three colors so that no two adjacent vertices have the same color. Use induction and Exercise 23 in Section 5.2. Finally, put guards at all vertices that are colored red, where red is the color used least in the coloring of the vertices. Show that placing guards at these points is all that is needed.]

Key Terms and Results

TERMS

undirected edge: an edge associated to a set $\{u, v\}$, where u and v are vertices

directed edge: an edge associated to an ordered pair (u, v) , where u and v are vertices

multiple edges: distinct edges connecting the same vertices

multiple directed edges: distinct directed edges associated with the same ordered pair (u, v) , where u and v are vertices

loop: an edge connecting a vertex with itself

undirected graph: a set of vertices and a set of undirected edges each of which is associated with a set of one or two of these vertices

simple graph: an undirected graph with no multiple edges or loops

multigraph: an undirected graph that may contain multiple edges but no loops

pseudograph: an undirected graph that may contain multiple edges and loops

directed graph: a set of vertices together with a set of directed edges each of which is associated with an ordered pair of vertices

directed multigraph: a graph with directed edges that may contain multiple directed edges

simple directed graph: a directed graph without loops or multiple directed edges

adjacent: two vertices are adjacent if there is an edge between them

incident: an edge is incident with a vertex if the vertex is an endpoint of that edge

deg v (degree of the vertex v in an undirected graph): the number of edges incident with v with loops counted twice

deg⁻(v) (the in-degree of the vertex v in a graph with directed edges): the number of edges with v as their terminal vertex

deg⁺(v) (the out-degree of the vertex v in a graph with directed edges): the number of edges with v as their initial vertex

underlying undirected graph of a graph with directed edges: the undirected graph obtained by ignoring the directions of the edges

K_n (complete graph on n vertices): the undirected graph with n vertices where each pair of vertices is connected by an edge

bipartite graph: a graph with vertex set that can be partitioned into subsets V_1 and V_2 so that each edge connects a vertex in V_1 and a vertex in V_2 . The pair (V_1, V_2) is called a **bipartition** of V .

$K_{m,n}$ (complete bipartite graph): the graph with vertex set partitioned into a subset of m elements and a subset of n elements with two vertices connected by an edge if and only if one is in the first subset and the other is in the second subset

C_n (cycle of size n), $n \geq 3$: the graph with n vertices $v_1, v_2, \dots, v_n$ and edges $\{v_1, v_2\}, \{v_2, v_3\}, \dots, \{v_{n-1}, v_n\}, \{v_n, v_1\}$

W_n (wheel of size n), $n \geq 3$: the graph obtained from C_n by adding a vertex and edges from this vertex to the original vertices in C_n

Q_n (n -cube), $n \geq 1$: the graph that has the 2^n bit strings of length n as its vertices and edges connecting every pair of bit strings that differ by exactly one bit

matching in a graph G : a set of edges such that no two edges have a common endpoint

complete matching M from V_1 to V_2 : a matching such that every vertex in V_1 is an endpoint of an edge in M

maximum matching: a matching containing the most edges among all matchings in a graph

isolated vertex: a vertex of degree zero

pendant vertex: a vertex of degree one

regular graph: a graph where all vertices have the same degree

subgraph of a graph $G = (V, E)$: a graph (W, F) , where W is a subset of V and F is a subset of E

$G_1 \cup G_2$ (union of G_1 and G_2): the graph $(V_1 \cup V_2, E_1 \cup E_2)$, where $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$

adjacency matrix: a matrix representing a graph using the adjacency of vertices

incidence matrix: a matrix representing a graph using the incidence of edges and vertices

isomorphic simple graphs: the simple graphs $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ are isomorphic if there exists a one-to-one correspondence f from V_1 to V_2 such that $\{f(v_1), f(v_2)\} \in E_2$ if and only if $\{v_1, v_2\} \in E_1$ for all v_1 and v_2 in V_1

invariant for graph isomorphism: a property that isomorphic graphs either both have or both do not have

path from u to v in an undirected graph: a sequence of edges $e_1, e_2, \dots, e_n$, where e_i is associated to $\{x_i, x_{i+1}\}$ for $i = 0, 1, \dots, n$, where $x_0 = u$ and $x_{n+1} = v$

path from u to v in a graph with directed edges: a sequence of edges $e_1, e_2, \dots, e_n$, where e_i is associated to (x_i, x_{i+1}) for $i = 0, 1, \dots, n$, where $x_0 = u$ and $x_{n+1} = v$

simple path: a path that does not contain an edge more than once

circuit: a path of length $n \geq 1$ that begins and ends at the same vertex

connected graph: an undirected graph with the property that there is a path between every pair of vertices

cut vertex of G : a vertex v such that $G - v$ is disconnected

cut edge of G : an edge e such that $G - e$ is disconnected

nonseparable graph: a graph without a cut vertex

vertex cut of G : a subset V' of the set of vertices of G such that $G - V'$ is disconnected

$\kappa(G)$ (the vertex connectivity of G): the size of a smallest vertex cut of G

k -connected graph: a graph that has a vertex connectivity no smaller than k

edge cut of G : a set of edges E' of G such that $G - E'$ is disconnected

$\lambda(G)$ (the edge connectivity of G): the size of a smallest edge cut of G

connected component of a graph G : a maximal connected subgraph of G

strongly connected directed graph: a directed graph with the property that there is a directed path from every vertex to every vertex

strongly connected component of a directed graph G : a maximal strongly connected subgraph of G

Euler path: a path that contains every edge of a graph exactly once

Euler circuit: a circuit that contains every edge of a graph exactly once

Hamilton path: a path in a graph that passes through each vertex exactly once

Hamilton circuit: a circuit in a graph that passes through each vertex exactly once

weighted graph: a graph with numbers assigned to its edges

shortest-path problem: the problem of determining the path in a weighted graph such that the sum of the weights of the edges in this path is a minimum over all paths between specified vertices

traveling salesperson problem: the problem that asks for the circuit of shortest total length that visits every vertex of a weighted graph exactly once

planar graph: a graph that can be drawn in the plane with no crossings

regions of a representation of a planar graph: the regions the plane is divided into by the planar representation of the graph

elementary subdivision: the removal of an edge $\{u, v\}$ of an undirected graph and the addition of a new vertex w together with edges $\{u, w\}$ and $\{w, v\}$

homeomorphic: two undirected graphs are homeomorphic if they can be obtained from the same graph by a sequence of elementary subdivisions

graph coloring: an assignment of colors to the vertices of a graph so that no two adjacent vertices have the same color

chromatic number: the minimum number of colors needed in a coloring of a graph

RESULTS

The handshaking theorem: If $G = (V, E)$ be an undirected graph with m edges, then $2m = \sum_{v \in V} \deg(v)$.

Hall's marriage theorem: The bipartite graph $G = (V, E)$ with bipartition (V_1, V_2) has a complete matching from V_1 to V_2 if and only if $|N(A)| \geq |A|$ for all subsets A of V_1 .

There is an Euler circuit in a connected multigraph if and only if every vertex has even degree.

There is an Euler path in a connected multigraph if and only if at most two vertices have odd degree.

Dijkstra's algorithm: a procedure for finding a shortest path between two vertices in a weighted graph (see Section 10.6)

Euler's formula: $r = e - v + 2$ where r , e , and v are the number of regions of a planar representation, the number of edges, and the number of vertices, respectively, of a connected planar graph

Kuratowski's theorem: A graph is nonplanar if and only if it contains a subgraph homeomorphic to $K_{3,3}$ or K_5 . (Proof beyond the scope of this book.)

The four color theorem: Every planar graph can be colored using no more than four colors. (Proof far beyond the scope of this book!)

Review Questions

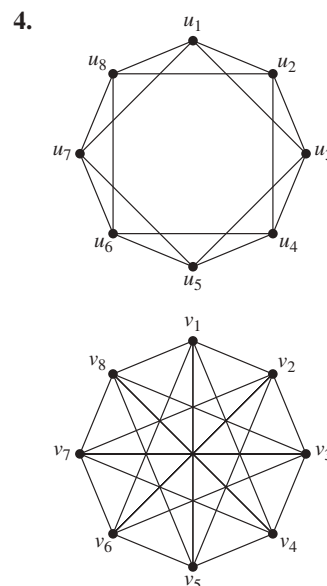
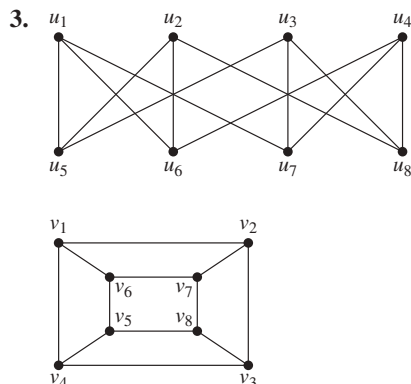
- Define a simple graph, a multigraph, a pseudograph, a directed graph, and a directed multigraph.
 - Use an example to show how each of the types of graph in part (a) can be used in modeling. For example, explain how to model different aspects of a computer network or airline routes.
- Give at least four examples of how graphs are used in modeling.
- What is the relationship between the sum of the degrees of the vertices in an undirected graph and the number of edges in this graph? Explain why this relationship holds.
- Why must there be an even number of vertices of odd degree in an undirected graph?
- What is the relationship between the sum of the in-degrees and the sum of the out-degrees of the vertices in a directed graph? Explain why this relationship holds.
- Describe the following families of graphs.
 - K_n , the complete graph on n vertices
 - $K_{m,n}$, the complete bipartite graph on m and n vertices
 - C_n , the cycle with n vertices
 - W_n , the wheel of size n
 - Q_n , the n -cube
- How many vertices and how many edges are there in each of the graphs in the families in Question 6?
- What is a bipartite graph?
 - Which of the graphs K_n , C_n , and W_n are bipartite?
 - How can you determine whether an undirected graph is bipartite?
- Describe three different methods that can be used to represent a graph.

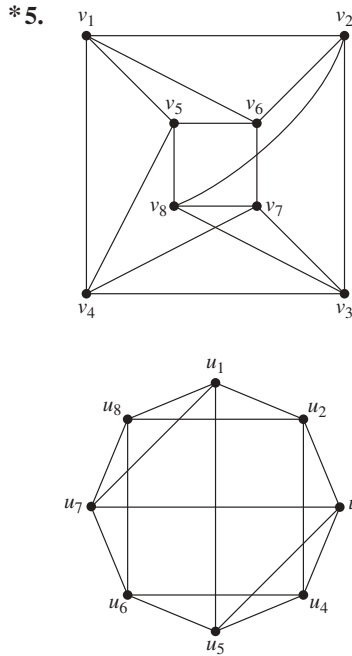
- b) Draw a simple graph with at least five vertices and eight edges. Illustrate how it can be represented using the methods you described in part (a).
10. a) What does it mean for two simple graphs to be isomorphic?
 b) What is meant by an invariant with respect to isomorphism for simple graphs? Give at least five examples of such invariants.
 c) Give an example of two graphs that have the same numbers of vertices, edges, and degrees of vertices, but that are not isomorphic.
 d) Is a set of invariants known that can be used to efficiently determine whether two simple graphs are isomorphic?
11. a) What does it mean for a graph to be connected?
 b) What are the connected components of a graph?
12. a) Explain how an adjacency matrix can be used to represent a graph.
 b) How can adjacency matrices be used to determine whether a function from the vertex set of a graph G to the vertex set of a graph H is an isomorphism?
 c) How can the adjacency matrix of a graph be used to determine the number of paths of length r , where r is a positive integer, between two vertices of a graph?
13. a) Define an Euler circuit and an Euler path in an undirected graph.
 b) Describe the famous Königsberg bridge problem and explain how to rephrase it in terms of an Euler circuit.
 c) How can it be determined whether an undirected graph has an Euler path?
- d) How can it be determined whether an undirected graph has an Euler circuit?
14. a) Define a Hamilton circuit in a simple graph.
 b) Give some properties of a simple graph that imply that it does not have a Hamilton circuit.
15. Give examples of at least two problems that can be solved by finding the shortest path in a weighted graph.
16. a) Describe Dijkstra's algorithm for finding the shortest path in a weighted graph between two vertices.
 b) Draw a weighted graph with at least 10 vertices and 20 edges. Use Dijkstra's algorithm to find the shortest path between two vertices of your choice in the graph.
17. a) What does it mean for a graph to be planar?
 b) Give an example of a nonplanar graph.
18. a) What is Euler's formula for connected planar graphs?
 b) How can Euler's formula for planar graphs be used to show that a simple graph is nonplanar?
19. State Kuratowski's theorem on the planarity of graphs and explain how it characterizes which graphs are planar.
20. a) Define the chromatic number of a graph.
 b) What is the chromatic number of the graph K_n when n is a positive integer?
 c) What is the chromatic number of the graph C_n when n is an integer greater than 2?
 d) What is the chromatic number of the graph $K_{m,n}$ when m and n are positive integers?
21. State the four color theorem. Are there graphs that cannot be colored with four colors?
22. Explain how graph coloring can be used in modeling. Use at least two different examples.

Supplementary Exercises

1. How many edges does a 50-regular graph with 100 vertices have?
2. How many nonisomorphic subgraphs does K_3 have?

In Exercises 3–5 determine whether two given graphs are isomorphic.





The **complete m -partite graph** $K_{n_1, n_2, \dots, n_m}$ has vertices partitioned into m subsets of $n_1, n_2, \dots, n_m$ elements each, and vertices are adjacent if and only if they are in different subsets in the partition.

6. Draw these graphs.

- a) $K_{1,2,3}$ b) $K_{2,2,2}$ c) $K_{1,2,2,3}$

*7. How many vertices and how many edges does the complete m -partite graph $K_{n_1, n_2, \dots, n_m}$ have?

8. Prove or disprove that there are always two vertices of the same degree in a finite multigraph having at least two vertices.

9. Let $G = (V, E)$ be an undirected graph and let $A \subseteq V$ and $B \subseteq V$. Show that

- a) $N(A \cup B) = N(A) \cup N(B)$.
 b) $N(A \cap B) \subseteq N(A) \cap N(B)$, and give an example where $N(A \cap B) \neq N(A) \cap N(B)$.

10. Let $G = (V, E)$ be an undirected graph. Show that

- a) $|N(v)| \leq \deg(v)$ for all $v \in V$.
 b) $|N(v)| = \deg v$ for all $v \in V$ if and only if G is a simple graph.

Suppose that $S_1, S_2, \dots, S_n$ is a collection of subsets of a set S where n is a positive integer. A **system of distinct representatives (SDR)** for this family is an ordered n -tuple $(a_1, a_2, \dots, a_n)$ with the property that $a_i \in S_i$ for $i = 1, 2, \dots, n$ and $a_i \neq a_j$ for all $i \neq j$.

11. Find a SDR for the sets $S_1 = \{a, c, m, e\}$, $S_2 = \{m, a, c, e\}$, $S_3 = \{a, p, e, x\}$, $S_4 = \{x, e, n, a\}$, $S_5 = \{n, a, m, e\}$, and $S_6 = \{e, x, a, m\}$.

12. Use Hall's marriage theorem to show that a collection of finite subsets $S_1, S_2, \dots, S_n$ of a set S has a SDR $(a_1, a_2, \dots, a_n)$ if and only if $|\bigcup_{i \in I} S_i| \geq |I|$ for all subsets I of $\{1, 2, \dots, n\}$.

13. a) Use Exercise 12 to show that the collection of sets $S_1 = \{a, b, c\}$, $S_2 = \{b, c, d\}$, $S_3 = \{a, b, d\}$, $S_4 = \{b, c, d\}$ has a SDR without finding one explicitly.

b) Find a SDR for the family of four sets in part (a).

14. Use Exercise 12 to show that collection of sets $S_1 = \{a, b, c\}$, $S_2 = \{a, c\}$, $S_3 = \{c, d, e\}$, $S_4 = \{b, c\}$, $S_5 = \{d, e, f\}$, $S_6 = \{a, c, e\}$, and $S_7 = \{a, b\}$ does not have a SDR.

The **clustering coefficient** $C(G)$ of a simple graph G is the probability that if u and v are neighbors and v and w are neighbors, then u and w are neighbors, where u , v , and w are distinct vertices of G .

15. We say that three vertices u , v , and w of a simple graph G form a triangle if there are edges connecting all three pairs of these vertices. Find a formula for $C(G)$ in terms of the number of triangles in G and the number of paths of length two in the graph. [Hint: Count each triangle in the graph once for each order of three vertices that form it.]

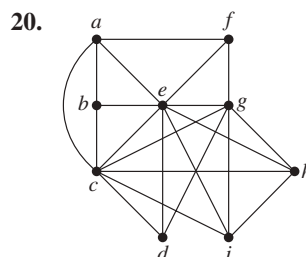
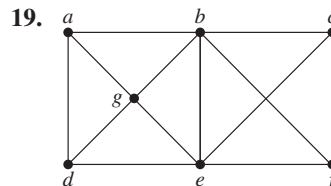
16. Find the clustering coefficient of each of the graphs in Exercise 20 of Section 10.2.

17. Explain what the clustering coefficient measures in each of these graphs.

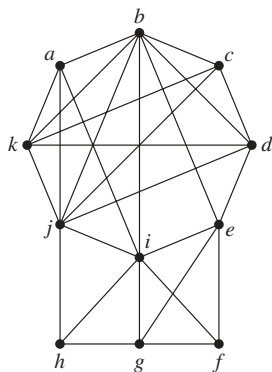
- a) the Hollywood graph
 b) the graph of Facebook friends
 c) the academic collaboration graph for researchers in graph theory
 d) the protein interaction graph for a human cell
 e) the graph representing the routers and communications links that make up the worldwide Internet

18. For each of the graphs in Exercise 17, explain whether you would expect its clustering coefficient to be closer to 0.01 or to 0.10 and why you expect this.

► A **clique** in a simple undirected graph is a complete subgraph that is not contained in any larger complete subgraph. In Exercises 19–21 find all cliques in the graph shown.

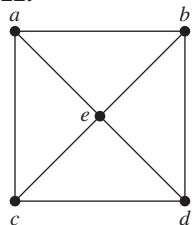


21.

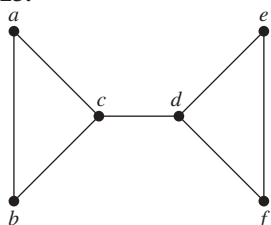


A **dominating set** of vertices in a simple graph is a set of vertices such that every other vertex is adjacent to at least one vertex of this set. A dominating set with the least number of vertices is called a **minimum dominating set**. In Exercises 22–24 find a minimum dominating set for the given graph.

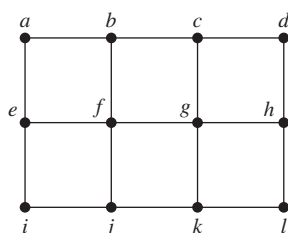
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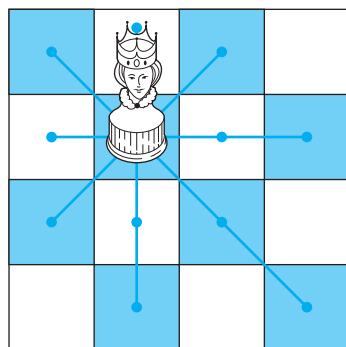
23.



24.



A simple graph can be used to determine the minimum number of queens on a chessboard that control the entire chessboard. An $n \times n$ chessboard has n^2 squares in an $n \times n$ configuration. A queen in a given position controls all squares in the same row, the same column, and on the two diagonals containing this square, as illustrated. The appropriate simple graph has n^2 vertices, one for each square, and two vertices are adjacent if a queen in the square represented by one of the vertices controls the square represented by the other vertex.



The squares controlled by a queen.

25. Construct the simple graph representing the $n \times n$ chessboard with edges representing the control of squares by queens for

- a) $n = 3$. b) $n = 4$.

26. Explain how the concept of a minimum dominating set applies to the problem of determining the minimum number of queens controlling an $n \times n$ chessboard.

**27. Find the minimum number of queens controlling an $n \times n$ chessboard for

- a) $n = 3$. b) $n = 4$. c) $n = 5$.

28. Suppose that G_1 and H_1 are isomorphic and that G_2 and H_2 are isomorphic. Prove or disprove that $G_1 \cup G_2$ and $H_1 \cup H_2$ are isomorphic.

29. Show that each of these properties is an invariant that isomorphic simple graphs either both have or both do not have.

- connectedness
- the existence of a Hamilton circuit
- the existence of an Euler circuit
- having crossing number C
- having n isolated vertices
- being bipartite

30. How can the adjacency matrix of $\overline{G}$ be found from the adjacency matrix of G , where G is a simple graph?

31. How many nonisomorphic connected bipartite simple graphs are there with four vertices?

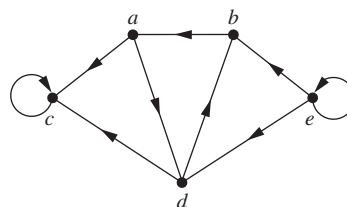
*32. How many nonisomorphic simple connected graphs with five vertices are there

- with no vertex of degree more than two?
- with chromatic number equal to four?
- that are nonplanar?

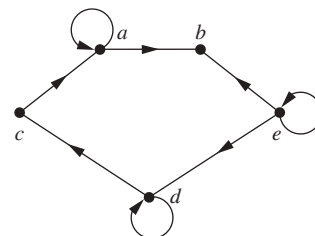
A directed graph is **self-converse** if it is isomorphic to its converse.

33. Determine whether the following graphs are self-converse.

a)

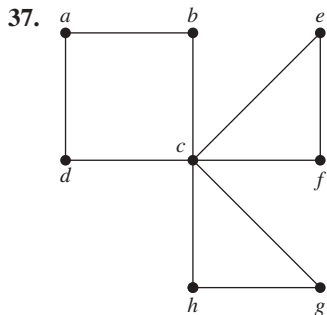
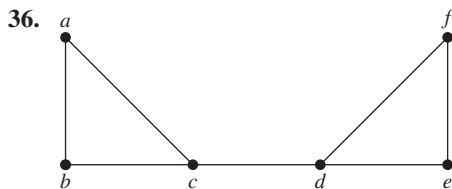
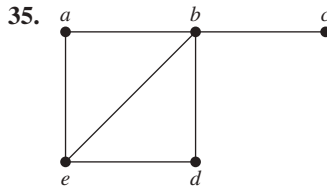


b)



34. Show that if the directed graph G is self-converse and H is a directed graph isomorphic to G , then H is also self-converse.

An **orientation** of an undirected simple graph is an assignment of directions to its edges such that the resulting directed graph is strongly connected. When an orientation of an undirected graph exists, this graph is called **orientable**. In Exercises 35–37 determine whether the given simple graph is orientable.



38. Because traffic is growing heavy in the central part of a city, traffic engineers are planning to change all the streets, which are currently two-way, into one-way streets. Explain how to model this problem.
- *39. Show that a graph is not orientable if it has a cut edge. A **tournament** is a simple directed graph such that if u and v are distinct vertices in the graph, exactly one of (u, v) and (v, u) is an edge of the graph.
40. How many different tournaments are there with n vertices?
41. What is the sum of the in-degree and out-degree of a vertex in a tournament?
- *42. Show that every tournament has a Hamilton path.
43. Given two chickens in a flock, one of them is dominant. This defines the **pecking order** of the flock. How can a tournament be used to model pecking order?

44. Suppose that a connected graph G has n vertices and vertex connectivity $\kappa(G) = k$. Show that G must have at least $\lceil kn/2 \rceil$ edges.

A connected graph $G = (V, E)$ with n vertices and m edges is said to have **optimal connectivity** if $\kappa(G) = \lambda(G) = \min_{v \in V} \deg v = 2m/n$.

45. Show that a connected graph with optimal connectivity must be regular.
46. Show these graphs have optimal connectivity.
- C_n for $n \geq 3$
 - K_n for $n \geq 3$
 - $K_{r,r}$ for $r \geq 2$
- *47. Find the two nonisomorphic simple graphs with six vertices and nine edges that have optimal connectivity.
48. Suppose that G is a connected multigraph with $2k$ vertices of odd degree. Show that there exist k subgraphs that have G as their union, where each of these subgraphs has an Euler path and where no two of these subgraphs have an edge in common. [Hint: Add k edges to the graph connecting pairs of vertices of odd degree and use an Euler circuit in this larger graph.]

In Exercises 49 and 50 we consider a puzzle posed by Petković in [Pe09] (based on a problem in [AvCh80]). Suppose that King Arthur has gathered his $2n$ knights of the Round Table for an important council. Every two knights are either friends or enemies, and each knight has no more than $n - 1$ enemies among the other $2n - 1$ knights. The puzzle asks whether King Arthur can seat his knights around the Round Table so that each knight has two friends for his neighbors.

49. a) Show that the puzzle can be reduced to determining whether there is a Hamilton circuit in the graph in which each knight is represented by a vertex and two knights are connected in the graph if they are friends.
b) Answer the question posed in the puzzle. [Hint: Use Dirac's theorem.]
50. Suppose that are eight knights Alynore, Bedivere, Degore, Gareth, Kay, Lancelot, Perceval, and Tristan. Their lists of enemies are A (D, G, P), B (K, P, T), D (A, G, L), G (A, D, T), K (B, L, P), L (D, K, T), P (A, B, K), T (B, G, L), where we have represented each knight by the first letter of his name and shown the list of enemies of that knight following this first letter. Draw the graph representing these eight knights and their friends and find a seating arrangement where each knight sits next to two friends.
- *51. Let G be a simple graph with n vertices. The **bandwidth** of G , denoted by $B(G)$, is the minimum, over all permutations $a_1, a_2, \dots, a_n$ of the vertices of G , of $\max\{|i - j| \mid a_i \text{ and } a_j \text{ are adjacent}\}$. That is, the bandwidth is the minimum over all listings of the vertices of the maximum difference in the indices assigned to adjacent vertices. Find the bandwidths of these graphs.
- K_5
 - $K_{1,3}$
 - $K_{2,3}$
 - $K_{3,3}$
 - Q_3
 - C_5

- *52. The **distance** between two distinct vertices v_1 and v_2 of a connected simple graph is the length (number of edges) of the shortest path between v_1 and v_2 . The **radius** of a graph is the minimum over all vertices v of the maximum distance from v to another vertex. The **diameter** of a graph is the maximum distance between two distinct vertices. Find the radius and diameter of
- a) K_6 . b) $K_{4,5}$. c) Q_3 . d) C_6 .
- *53. a) Show that if the diameter of the simple graph G is at least four, then the diameter of its complement $\bar{G}$ is no more than two.
b) Show that if the diameter of the simple graph G is at least three, then the diameter of its complement $\bar{G}$ is no more than three.
- *54. Suppose that a multigraph has $2m$ vertices of odd degree. Show that any circuit that contains every edge of the graph must contain at least m edges more than once.
55. Find the second shortest path between the vertices a and z in Figure 3 of Section 10.6.
56. Devise an algorithm for finding the second shortest path between two vertices in a simple connected weighted graph.
57. Find the shortest path between the vertices a and z that passes through the vertex f in the weighted graph in Exercise 3 in Section 10.6.
58. Devise an algorithm for finding the shortest path between two vertices in a simple connected weighted graph that passes through a specified third vertex.
- *59. Show that if G is a simple graph with at least 11 vertices, then either G or $\bar{G}$, the complement of G , is nonplanar.
- ▶ A set of vertices in a graph is called **independent** if no two vertices in the set are adjacent. The **independence number** of a graph is the maximum number of vertices in an independent set of vertices for the graph.
- *60. What is the independence number of
- a) K_n ? b) C_n ? c) Q_n ? d) $K_{m,n}$?
61. Show that the number of vertices in a simple graph is less than or equal to the product of the independence number and the chromatic number of the graph.
62. Show that the chromatic number of a graph is less than or equal to $n - i + 1$, where n is the number of vertices in the graph and i is the independence number of this graph.
63. Suppose that to generate a random simple graph with n vertices we first choose a real number p with $0 \leq p \leq 1$. For each of the $C(n, 2)$ pairs of distinct vertices we generate a random number x between 0 and 1. If $0 \leq x \leq p$, we connect these two vertices with an edge; otherwise these vertices are not connected.
- a) What is the probability that a graph with m edges where $0 \leq m \leq C(n, 2)$ is generated?
b) What is the expected number of edges in a randomly generated graph with n vertices if each edge is included with probability p ?
c) Show that if $p = 1/2$ then every simple graph with n vertices is equally likely to be generated.
- A property retained whenever additional edges are added to a simple graph (without adding vertices) is called **monotone increasing**, and a property that is retained whenever edges are removed from a simple graph (without removing vertices) is called **monotone decreasing**.
64. For each of these properties, determine whether it is monotone increasing and determine whether it is monotone decreasing.
- a) The graph G is connected.
b) The graph G is not connected.
c) The graph G has an Euler circuit.
d) The graph G has a Hamilton circuit.
e) The graph G is planar.
f) The graph G has chromatic number four.
g) The graph G has radius three.
h) The graph G has diameter three.
65. Show that the graph property P is monotone increasing if and only if the graph property Q is monotone decreasing where Q is the property of not having property P .
- **66. Suppose that P is a monotone increasing property of simple graphs. Show that the probability a random graph with n vertices has property P is a monotonic nondecreasing function of p , the probability an edge is chosen to be in the graph.

Computer Projects

Write programs with these input and output.

- Given the vertex pairs associated to the edges of an undirected graph, find the degree of each vertex.
- Given the ordered pairs of vertices associated to the edges of a directed graph, determine the in-degree and out-degree of each vertex.
- Given the list of edges of a simple graph, determine whether the graph is bipartite.
- Given the vertex pairs associated to the edges of a graph, construct an adjacency matrix for the graph. (Produce a version that works when loops, multiple edges, or directed edges are present.)
- Given an adjacency matrix of a graph, list the edges of this graph and give the number of times each edge appears.
- Given the vertex pairs associated to the edges of an undirected graph and the number of times each edge appears, construct an incidence matrix for the graph.

7. Given an incidence matrix of an undirected graph, list its edges and give the number of times each edge appears.
8. Given a positive integer n , generate a simple graph with n vertices by producing an adjacency matrix for the graph so that all simple graphs with n vertices are equally likely to be generated.
9. Given a positive integer n , generate a simple directed graph with n vertices by producing an adjacency matrix for the graph so that all simple directed graphs with n vertices are equally likely to be generated.
10. Given the lists of edges of two simple graphs with no more than six vertices, determine whether the graphs are isomorphic.
11. Given an adjacency matrix of a graph and a positive integer n , find the number of paths of length n between two vertices. (Produce a version that works for directed and undirected graphs.)
- *12. Given the list of edges of a simple graph, determine whether it is connected and find the number of connected components if it is not connected.
13. Given the vertex pairs associated to the edges of a multigraph, determine whether it has an Euler circuit and, if not, whether it has an Euler path. Construct an Euler path or circuit if it exists.
- *14. Given the ordered pairs of vertices associated to the edges of a directed multigraph, construct an Euler path or Euler circuit, if such a path or circuit exists.
- **15. Given the list of edges of a simple graph, produce a Hamilton circuit, or determine that the graph does not have such a circuit.
- **16. Given the list of edges of a simple graph, produce a Hamilton path, or determine that the graph does not have such a path.
17. Given the list of edges and weights of these edges of a weighted connected simple graph and two vertices in this graph, find the length of a shortest path between them using Dijkstra's algorithm. Also, find a shortest path.
18. Given the list of edges of an undirected graph, find a coloring of this graph using the algorithm given in the exercise set of Section 10.8.
19. Given a list of students and the courses that they are enrolled in, construct a schedule of final exams.
20. Given the distances between pairs of television stations and the minimum allowable distance between stations, assign frequencies to these stations.

Computations and Explorations

Use a computational program or programs you have written to do these exercises.

1. Display all simple graphs with four vertices.
2. Display a full set of nonisomorphic simple graphs with six vertices.
3. Display a full set of nonisomorphic directed graphs with four vertices.
4. Generate at random 10 different simple graphs each with 20 vertices so that each such graph is equally likely to be generated.
5. Construct a Gray code where the code words are bit strings of length six.
6. Construct knight's tours on chessboards of various sizes.
7. Determine whether each of the graphs you generated in Exercise 4 of this set is planar. If you can, determine the thickness of each of the graphs that are not planar.
8. Determine whether each of the graphs you generated in Exercise 4 of this set is connected. If a graph is not connected, determine the number of connected components of the graph.
9. Generate at random simple graphs with 10 vertices. Stop when you have constructed one with an Euler circuit. Display an Euler circuit in this graph.
10. Generate at random simple graphs with 10 vertices. Stop when you have constructed one with a Hamilton circuit. Display a Hamilton circuit in this graph.
11. Find the chromatic number of each of the graphs you generated in Exercise 4 of this set.
- **12. Find the shortest path a traveling salesperson can take to visit each of the capitals of the 50 states in the United States, traveling by air between cities in a straight line.
- *13. Estimate the probability that a randomly generated simple graph with n vertices is connected for each positive integer n not exceeding ten by generating a set of random simple graphs and determining whether each is connected.
- **14. Work on the problem of determining whether the crossing number of $K_{7,7}$ is 77, 79, or 81. It is known that it equals one of these three values.

Writing Projects

Respond to these with essays using outside sources.

1. Describe the origins and development of graph theory prior to the year 1900.
2. Discuss the applications of graph theory to the study of ecosystems.
3. Discuss the applications of graph theory to sociology and psychology.
4. Discuss what can be learned by investigating the properties of the web graph.
5. Explain what community structure is in a graph representing a network, such as a social network, a computer network, an information network, or a biological network. Define what a community in such a graph is, and explain what communities represent in graphs representing the types of networks listed.
6. Describe some of the algorithms used to detect communities in graphs representing networks of the types listed in Question 5.
7. Describe algorithms for drawing a graph on paper or on a display given the vertices and edges of the graph. What considerations arise in drawing a graph so that it has the best appearance for understanding its properties?
8. Explain how graph theory can help uncover networks of criminals or terrorists by studying relevant social and communication networks.
9. What are some of the capabilities that a software tool for inputting, displaying, and manipulating graphs should have? Which of these capabilities do available tools have?
10. Describe some of the algorithms available for determining whether two graphs are isomorphic and the computational complexity of these algorithms. What is the most efficient such algorithm currently known?
11. What is the subgraph isomorphism problem and what are some of its important applications, including those to chemistry, bioinformatics, electronic circuit design, and computer vision?
12. Explain what the area of graph mining, an important area of data mining, is and describe some of the basic techniques used in graph mining.
13. Describe how Euler paths can be used to help determine DNA sequences.
14. Define *de Bruijn sequences* and discuss how they arise in applications. Explain how de Bruijn sequences can be constructed using Euler circuits.
15. Describe the *Chinese postman problem* and explain how to solve this problem.
16. Describe some of the different conditions that imply that a graph has a Hamilton circuit.
17. Describe some of the strategies and algorithms used to solve the traveling salesperson problem.
18. Describe several different algorithms for determining whether a graph is planar. What is the computational complexity of each of these algorithms?
19. In modeling, very large scale integration (VLSI) graphs are sometimes embedded in a book, with the vertices on the spine and the edges on pages. Define the *book number* of a graph and find the book number of various graphs including K_n for $n = 3, 4, 5$, and 6 .
20. Discuss the history of the four color theorem.
21. Describe the role computers played in the proof of the four color theorem. How can we be sure that a proof that relies on a computer is correct?
22. Describe and compare several different algorithms for coloring a graph, in terms of whether they produce a coloring with the least number of colors possible and in terms of their complexity.
23. Explain how graph multicolorings can be used in a variety of different models.
24. Describe some of the applications of edge colorings.
25. Explain how the theory of random graphs can be used in nonconstructive existence proofs of graphs with certain properties.

